

INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

A Project Report

On

SUPPLY CHAIN VISIBILITY SYSTEM WITH OPTIMIZATION ANALYTICS

Milestone - 1 Documentation: Data Modeling & KPI
Foundation

Submitted By-

Varun Singh

1. INTRODUCTION

The **Supply Chain Visibility System with Optimization Analytics** is a data analytics project developed as part of the Infosys Springboard Virtual Internship. The main purpose of the project is to analyze vessel movement and logistics-related data and convert the available raw data into meaningful information through data preparation, modeling, KPI development, and interactive dashboards.

In Milestone 1, the focus was on establishing the basic foundation of the project. The dataset was imported into Power BI, examined to understand its structure and important attributes, and prepared for analysis. Data cleaning and transformation were performed wherever required, followed by the creation of a simple data model and relationship between the required tables. DAX measures were then developed to calculate important KPIs, which were used to create an interactive dashboard for understanding vessel activity, speed, distance, ETA, vessel type, and status.

1.1 MILESTONE 1 REQUIREMENTS

The major requirements completed during Milestone 1 were:

- **Data Import:** Import the provided logistics/vessel movement dataset into Power BI.
- **Data Preparation:** Examine the dataset structure, identify relevant columns, and prepare the data for analysis.
- **Data Cleaning:** Handle data types, unnecessary inconsistencies, and prepare columns required for analysis.
- **Data Modeling:** Create an appropriate data model using the required tables and establish relationships between them.
- **DAX Development:** Create DAX measures and calculated columns required for analytical calculations.
- **KPI Development:** Develop important KPIs such as total unique vessels, total vessel records, average speed, and average ETA.
- **Dashboard Development:** Build an interactive dashboard containing KPI cards, charts, slicers, and a detailed vessel information table.
- **Data Visualization:** Present the analyzed information in a clean and easy-to-understand visual format.

1.2 MILESTONE 1 OBJECTIVE

- The overall objective of Milestone 1 was to build a strong analytical foundation for the Supply Chain Visibility System, so that the prepared data, data model, KPIs, and dashboard can be further extended in the upcoming milestones.

2. DATASET OVERVIEW

The dataset used in this project contains vessel movement and maritime logistics information. It records different observations related to vessels, including their identification details, location, movement characteristics, cargo information, estimated arrival time, vessel type, status, and weather conditions. The dataset provides the basic information required to understand vessel activity and develop supply-chain visibility insights.

The dataset contains a large number of vessel movement records, which makes it suitable for analyzing vessel behavior and operational patterns. Each record represents an observation of a vessel at a particular time, allowing different parameters such as speed, location, distance, ETA, and vessel status to be analyzed together.

2.1 MAJOR CATEGORIES OF DATA

The dataset can be broadly divided into the following categories:

VESSEL DATA CATEGORY BREAKDOWN		
Data Category	Examples of Fields	Purpose
 Vessel Identification	MMSI, VesselName, CallSign, IMO	Identifies individual vessels
 Time Information	BaseDateTime, MovementDate, Hour	Helps analyze vessel movement over time
 Location Information	LAT, LON, destination latitude/longitude	Represents vessel position and destination
 Movement Information	SOG, COG, SOG_kmh, Speed_Category	Describes vessel movement and speed
 Operational Information	Cargo, VesselType, Status	Provides information about vessel activity
 ETA Information	ETA_hours, ETA_min	Helps analyze estimated arrival time
 Weather Information	Visibility, Cloud, Humidity, Pressure, Temperature, Precipitation	Provides environmental conditions associated with vessel movement

Source Data: Standard Maritime Reporting

2.2 IMPORTANT DATASET FIELDS

Some of the important fields used during Milestone 1 are:

- **MMSI:** A unique maritime identification number used to identify a vessel.
- **VesselName:** Name of the vessel.
- **BaseDateTime:** Date and time at which the vessel observation was recorded.
- **LAT:** Latitude of the vessel's recorded position.
- **LON:** Longitude of the vessel's recorded position.
- **SOG:** Speed over ground, representing the vessel's movement speed.
- **SOG_kmh:** Speed converted into kilometers per hour.
- **Speed_Category:** A categorized representation of vessel speed such as *Stopped*, *Slow*, *Moderate*, and *Fast*.
- **Cargo:** Represents the cargo-related value associated with the vessel record.
- **VesselType:** Represents the category/type of vessel.
- **Status:** Represents the operational/status value recorded for the vessel.
- **ETA_hours:** Estimated time of arrival represented in hours.
- **MovementDate:** Date extracted from the vessel movement timestamp for date-based analysis.
- **Hour / Hour_Label:** Additional time fields created to analyze vessel activity and ETA patterns at different hours.
- **Weather Fields:** These include environmental parameters such as visibility, cloud conditions, humidity, pressure, temperature, and precipitation.

2.3 DATASET UNDERSTANDING

During the initial examination of the dataset, we identified that it contains both raw operational information and fields that can be transformed into analytical features. For example, the original timestamp can be used to create date and hour-based fields, while vessel speed can be grouped into meaningful speed categories.

This understanding helped us decide which fields were directly useful for analysis and which additional fields needed to be created during the data preparation stage. The prepared dataset was then used for data modeling, DAX calculations, KPI development, and dashboard creation.

3. TOOLS & TECHNOLOGIES

The development of the Supply Chain Visibility System was carried out mainly using Microsoft Power BI and its integrated data analysis tools. Different components of Power BI were used for importing, preparing, transforming, modeling, calculating, and visualizing the vessel movement data.

Tool / Technology	Purpose in the Project
Microsoft Power BI Desktop	Main platform used to develop the data model, KPIs, visualizations, and interactive dashboard
Power Query	Used for importing, transforming, and preparing the raw dataset before analysis
DAX (Data Analysis Expressions)	Used to create calculated columns and measures for KPI and analytical calculations
Power BI Data Model	Used to organize the tables and establish the required relationship between the date and vessel movement data
Power BI Visualizations	Used to present the analyzed information through KPI cards, charts, slicers, and tables

Table view from [Project Name]

4. DATA IMPORT

The first step in developing the Supply Chain Visibility System was to import the provided vessel movement dataset into Microsoft Power BI Desktop. Since the project involves a large number of vessel movement records, Power BI was selected as the main platform for handling, analyzing, and visualizing the data.

The dataset was imported into Power BI and loaded into the Power Query Editor, where the structure and contents of the data could be examined before creating the final analytical model. During this initial stage, the available columns were reviewed to understand the type of information contained in the dataset and to identify fields that would be useful for further analysis.

4.1 DATASET IMPORT PROCESS

The data import process was carried out through the following steps:

1. Power BI Desktop was opened and a new project was created.
2. The Get Data option was used to connect Power BI with the provided dataset.
3. The required dataset was selected and loaded into Power Query.

4. The available columns and sample records were inspected to understand the structure of the data.
5. Data types of important fields were reviewed, especially fields related to dates, vessel identification, location, speed, cargo, ETA, and vessel type.
6. The imported query was organized and renamed as Fact_VesselMovement to represent the main vessel movement table in the data model.
7. The data was then prepared for the cleaning and transformation stage.

4.2 UNDERSTANDING THE IMPORTED DATA

After importing the dataset, we identified that the main table contained information related to vessel identification, movement, geographical location, cargo, vessel type, status, ETA, and weather conditions.

The BaseDateTime field was particularly important because it contained the date and time associated with vessel observations. This field later helped us create additional time-related fields such as BaseDate, MovementDate, Hour, and Hour_Label, which were required for time-based analysis in the dashboard.

Similarly, fields such as MMSI, SOG, LAT, LON, Cargo, VesselType, Status, and ETA_hours were reviewed because they could directly support the KPIs and visualizations planned for Milestone 1.

4.3 PREPARING THE DATA FOR THE NEXT STAGE

The imported data was not immediately used for dashboard development. First, it was taken through a data preparation and cleaning process to make the information suitable for analysis.

This ensured that the dataset had appropriate data types, useful analytical fields, and a consistent structure before moving to the modeling stage. After completing the initial import and inspection, the next step was therefore to clean and transform the dataset using Power Query.

5. DATA PREPARATION AND CLEANING

After importing the dataset, the next step was to prepare the raw vessel movement data for analysis. Data preparation was important because the raw dataset contained different types of information, and the fields needed to be reviewed and organized before they could be used effectively for calculations and visualizations.

The data preparation process was mainly carried out using Power Query in Power BI. The objective was to make sure that the important fields had appropriate data types, identify fields required for analysis, and create additional fields that could provide more useful analytical information.

5.1 DATA TYPE VALIDATION

The data types of the important columns were reviewed during the preparation stage. Correct data types are important because Power BI uses them differently during calculations, filtering, sorting, and visualization.

The following types of fields were checked:

- **Vessel identification fields** such as MMSI and IMO were treated as identification values rather than numerical measurements.
- **Date and time fields** were reviewed so they could be used for date and time analysis.
- **Latitude and longitude** fields were maintained as numerical values for geographical analysis.
- **Speed-related fields** were checked as numerical values so that averages and categories could be analyzed.
- **ETA and distance fields** were reviewed for numerical calculations.
- **Categorical fields** such as vessel status, vessel type and cargo were prepared for filtering and classification.

5.2 DATA PREPARATION

The following fields were created:

Columns Created During Data Preparation

Column Created	Purpose
BaseDate	Extracts only the date from BaseDateTime for date-based analysis and for connecting the fact table with the DimDate table.
Hour	Extracts the hour from BaseDateTime to support hourly analysis and provide a numeric field for correct chart sorting.
Hour_Label	Converts the hour into a readable format such as 00:00, 01:00 and 02:00 for clear display on dashboard charts.

6. DATA MODELING & RELATIONSHIP CREATION

After completing the data preparation and cleaning process, the next step was to create a structured data model in Power BI. Data modeling was required to organize the prepared data and establish a connection between the vessel movement records and the date information used for time-based analysis.

The model developed for Milestone 1 consists mainly of two tables: **Fact_VesselMovement** and **DimDate**. The **Fact_VesselMovement** table contains the detailed vessel movement records, while the **DimDate** table provides a structured date reference for date-based filtering and analysis.

6.1 FACT_VESSELMOVEMENT TABLE

The **Fact_VesselMovement** table was used as the main table of the data model. It contains the detailed records imported and prepared during the earlier stages.

The table contains information related to:

- Vessel identification
- Date and time of movement
- Vessel location
- Speed and speed category
- Vessel type
- Vessel status
- Cargo information
- Distance
- ETA
- Other operational and weather-related fields

Since multiple movement records can exist for the same vessel and date, this table represents the detailed movement-level data used for calculations and dashboard visualizations.

6.2 DIMDATE TABLE

A separate **DimDate** table was created to provide a dedicated date structure for the project.

Instead of depending completely on the date values stored within the movement table, the date dimension provides a consistent source for date-based analysis and filtering.

The DimDate table was connected to the prepared date field in the Fact_VesselMovement table. This structure also provides a foundation for future time-based analysis as the project is expanded in later milestones.

6.3 RELATIONSHIP CREATION

A **one-to-many (1:*) relationship** was created between the two tables.

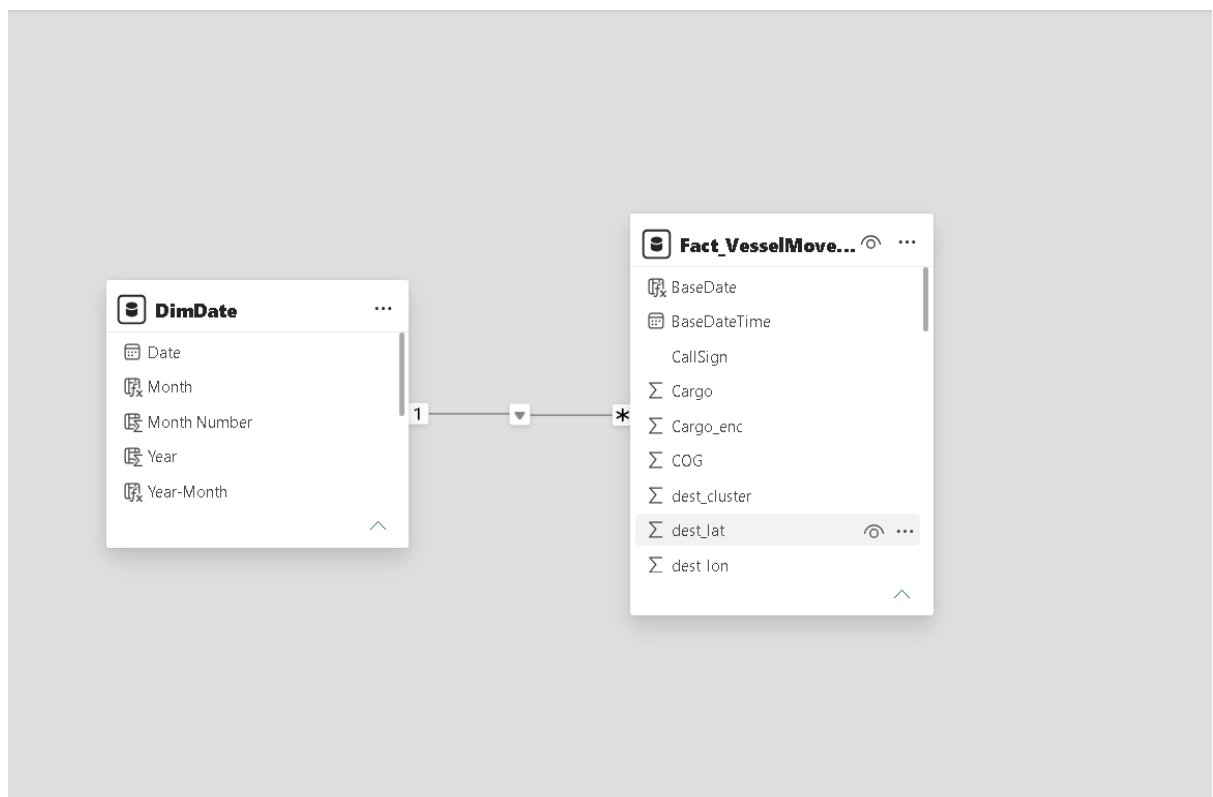
The relationship can be represented as:

DimDate (1) → Fact_VesselMovement (*)

The **DimDate** table is placed on the one side because each date is represented once in the date dimension. On the other hand, the **Fact_VesselMovement** table is placed on the many side because multiple vessel movement records can belong to the same date.

The relationship was created using the appropriate date fields from both tables. It was kept active so that date-based selections and filters could be applied to the vessel movement data.

This relationship allows the dashboard to use the date dimension for analysis while the detailed movement records remain in the main fact table.



7. DAX DEVELOPMENT

After creating the data model, the next step was to develop the calculations required for analysis. Data Analysis Expressions (DAX) was used in Power BI to create calculated columns and measures from the prepared vessel movement data.

DAX was important because the raw dataset alone does not directly provide all the values required for the dashboard. By using DAX, the required information could be calculated dynamically and then used in KPI cards and other visualizations.

7.1 DAX CALCULATED COLUMNS

During the data preparation stage, additional time-related columns were created to support the dashboard's time-based analysis.

The main calculated columns were:

- **BaseDate** — extracts the date from BaseDateTime and is used for date-level analysis.
- **Hour** — extracts the hour from the movement timestamp and is used for hourly analysis.
- **Hour_Label** — provides a readable hour representation for displaying hourly information in charts.

7.2 DAX MEASURES

DAX measures were then created to calculate the main numerical indicators required for Milestone 1. The measures were designed to provide summarized information from the vessel movement data. Unlike fixed values, these measures are calculated dynamically based on the underlying data and can respond to filters applied through the dashboard.

The major measures developed were:

Key DAX Measures Used in the Dashboard

Measure	DAX Formula	Purpose
Total Unique Vessels	DISTINCTCOUNT(Fact_VesselMovement[MMSI])	Counts the distinct vessels represented in the dataset.
Total Vessel Records	COUNTROWS(Fact_VesselMovement)	Counts the total number of vessel movement records.
Average Speed	AVERAGE(Fact_VesselMovement[SOG_kmh])	Calculates the average vessel speed in km/h.
Average ETA	AVERAGE(Fact_VesselMovement[ETA_hours])	Calculates the average estimated time of arrival in hours.
Average Distance	AVERAGE(Fact_VesselMovement[dist_km])	Calculates the average distance used in dashboard analysis.
DAX measures used to calculate the key dashboard indicators		

8. KPI DEVELOPMENT

After creating the required DAX calculations, the next step was to develop the Key Performance Indicators (KPIs) for the dashboard. KPIs were created to provide a quick and clear summary of important vessel movement information without requiring the user to examine individual records.

Four main KPI cards were included in the final dashboard:

8.1 TOTAL UNIQUE VESSELS

This KPI represents the number of distinct vessels available in the dataset. It provides a quick understanding of the overall number of individual vessels represented by the movement records.

8.2 TOTAL VESSEL RECORDS

This KPI represents the total number of vessel movement records available in the dataset. Since a single vessel can have multiple movement observations, this value provides an overview of the overall volume of recorded vessel activity.

8.3 AVERAGE SPEED

The Average Speed KPI provides the overall average movement speed of the vessels. It helps give a quick indication of the general movement condition represented in the dataset.

8.4 AVERAGE ETA

The Average ETA KPI represents the average estimated time of arrival based on the available vessel records. It provides a summarized view of the expected arrival information.

MARITIME KPI DASHBOARD				
KPI		What It Represents	Purpose	
Total Unique Vessels		Number of distinct vessels	Provides an overview of vessel count	
Total Vessel Records		Total movement observations	Shows the overall volume of vessel activity	
Average Speed		Average vessel movement speed	Provides an overall view of vessel movement	
Average ETA		Average estimated arrival time	Provides a summarized view of arrival information	

9. DASHBOARD DEVELOPMENT

After completing data preparation, data modeling, DAX calculations, and KPI development, the final step of Milestone 1 was to develop an interactive Power BI dashboard. The dashboard was designed to present the important vessel movement information in a clear and organized manner so that users can understand the overall situation without going through the complete dataset manually.

The dashboard combines KPI cards, analytical charts, a detailed data table, and an interactive slicer. The layout was designed to keep the most important summary information at the top while providing more detailed analysis through the visualizations below.

9.1 DASHBOARD LAYOUT AND DESIGN

The dashboard was arranged into different sections according to the type of information being presented.

The top section contains the main KPI cards, which provide an immediate summary of the vessel movement data. The analytical charts are placed below the KPIs so that users can move from a high-level overview to more detailed analysis.

The dashboard was also given a consistent visual theme with a suitable background, colored KPI cards, and formatted visual elements. The objective was to make the dashboard both visually appealing and easy to read, while avoiding unnecessary visual clutter.

9.2 KPI CARDS

Four KPI cards were placed prominently on the dashboard:

- **Total Unique Vessels**
- **Total Vessel Records**
- **Average Speed**
- **Average ETA**

These cards provide the first level of information to the user. Instead of examining the complete dataset, users can immediately see the main numerical indicators.

The KPI values are generated using the DAX measures developed earlier, which allows them to respond to the data and applied filters.

9.3 ANALYTICAL VISUALIZATIONS

Several charts were included to analyze different aspects of vessel movement.

Speed Category Distribution

This visual shows the distribution of vessel movement records across different speed categories. It provides an easy way to understand how the recorded vessels are distributed according to their movement speed.

Average Distance by Status

This chart compares the average distance associated with different vessel status categories. It helps in understanding how distance varies according to the operational status of the vessels.

Average ETA by Hour

This visualization uses the time-related fields created during data preparation. It shows how the average estimated time of arrival changes across different hours.

This visual demonstrates the usefulness of the Hour and Hour_Label fields created during the data preparation stage.

Vessel Type Distribution

This chart provides an overview of the different vessel types represented in the dataset. It helps users understand the composition of the vessel records.

9.4 VESSEL DETAILS TABLE

A detailed table was included in the dashboard to complement the graphical visualizations.

While charts provide summarized information, the table allows users to inspect individual vessel records and their associated information. This provides a more detailed view when a user needs to examine specific records rather than only overall trends.

9.5 INTERACTIVE VESSEL STATUS SLICER

A Vessel Status slicer was included to make the dashboard interactive.

The slicer allows users to select a particular vessel status and analyze the dashboard according to that selection. When a status is selected, the relevant visualizations and KPI values can respond to the applied filter.

This makes the dashboard more useful than a static collection of charts because users can interact with the data and focus on the information that is relevant to them.

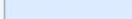
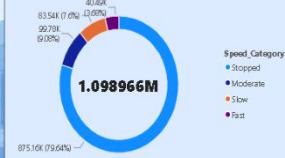
Vessel Movement & Supply Chain Visibility Dashboard

Total Unique Vessels

Total Vessel Records

Average Speed

Average ETA



Status

☐ 0

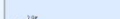
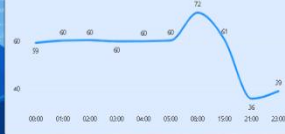
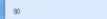
5

□ 3

□

Vess

MMSI	Cargo	LAT	LO
477321000	COAL	37.15	122.25

[illegible]

MILESTONE 2 - INVENTORY & DELIVERY ANALYTICS

10.1. INTRODUCTION

Following the completion of Milestone 1, the existing data model and KPI foundation were further extended in Milestone 2 to perform deeper operational analysis. This milestone focuses on analyzing vessel movement, destination, cargo, distance, ETA, and variance-related information using Power BI.

The dashboard was enhanced with additional KPI measures, analytical charts, drill-down analysis, trend and variance analysis, and interactive filtering. These enhancements provide deeper insights into operational patterns and support more detailed analysis of vessel movement and arrival performance.

10.2. MILESTONE 2 REQUIREMENTS & OBJECTIVES

10.2.1 REQUIREMENTS

The main requirements for Milestone 2 were to extend the Power BI dashboard with deeper inventory, delivery, drill-down, trend, and variance analysis. The suggested visualizations included:

- **KPI Cards** for important performance indicators.
- **Bar Chart** for category-wise analysis.
- **Column Chart** for warehouse or location-wise analysis.
- **Line Chart** for analyzing trends over time.
- **Table** for detailed record-level information.
- **Donut Chart** for comparing on-time and delayed deliveries.
- **Drill-down charts** for hierarchical analysis.
- **Trend and variance visuals** to identify changes and differences in performance.

10.2.2 OBJECTIVES

The objectives of Milestone 2 were:

1. To enhance the existing Power BI dashboard with deeper operational analysis.
2. To analyze vessel movement, destination, cargo, distance, and ETA-related performance.

3. To identify slow-moving and fast-moving vessel records using KPI measures.
4. To analyze movement and ETA trends across different time periods and destinations.
5. To identify variations in ETA performance using variance analysis.
6. To provide drill-down capability from Cargo to Vessel for detailed analysis.
7. To provide interactive filtering through the Destination Cluster slicer.
8. To present the analysis through clear and visually understandable Power BI charts.

10.3. DATA & EXISTING MODEL

Milestone 2 uses the existing data model developed during Milestone 1. The model is designed to organize vessel movement and date-related information for analysis in Power BI. No separate data model was created for Milestone 2; instead, the existing model was extended through additional DAX measures and analytical visuals.

10.3.1 DIMDATE TABLE

The **DimDate** table provides the date-related fields required for time-based analysis. It contains:

- Date
- Year
- Month
- Month Number
- Year-Month

The table is used to support date and time-based analysis and maintain consistency across the dashboard.

10.3.2 FACT_VESSELMOVEMENT TABLE

The **Fact_VesselMovement** table contains the main vessel movement and operational data used throughout the dashboard. Important fields used in Milestone 2 include:

Important Fields Used in Milestone 2

Field	Purpose
MMSI	Identifies vessel records
VesselName	Identifies individual vessels
Cargo	Supports cargo-wise analysis
dest_cluster	Supports destination-wise analysis
dist_km	Used for distance analysis
ETA_hours	Used for ETA and variance analysis
SOG_kmh	Used for vessel speed analysis
Speed_Category	Identifies movement speed categories

10.3.3 MODEL UTILIZATION IN MILESTONE 2

The existing model was used to create new analytical measures and visualizations for:

- Destination and distance analysis
- ETA analysis
- Vessel movement trends
- Cargo analysis
- Drill-down from **Cargo** → **Vessel**
- Variance analysis
- Distance and ETA relationship analysis

This allowed Milestone 2 to build upon the foundation created in Milestone 1 without duplicating the complete data preparation and modelling process.

10.4. DAX MEASURES DEVELOPMENT

To support the analytical requirements of Milestone 2, five new DAX measures were developed in Power BI using the existing Fact_VesselMovement table. These measures were used to create KPI cards and analytical visuals for distance, vessel speed, ETA variability, and destination-wise ETA analysis.

10.4.1 TOTAL DISTANCE TRAVELLED (KM)

Purpose:

This measure calculates the total recorded distance travelled by all vessel movement records.

DAX Formula:

Total Distance Travelled (km) =
`SUM('Fact_VesselMovement'[dist_km])`

Explanation:

The SUM function adds all the values from the dist_km column to calculate the overall distance travelled. The resulting value is used as an important KPI to understand the total movement distance represented in the dataset.

Used in:

- KPI Card – **Total Distance Travelled (km)**
- Destination-wise distance analysis

10.4.2 SLOW-MOVING VESSELS

Purpose:

This measure calculates the number of unique vessels classified as **Slow** according to the Speed_Category field.

DAX Formula:

Slow-Moving Vessels =
`CALCULATE(
 DISTINCTCOUNT('Fact_VesselMovement'[MMSI]),
 'Fact_VesselMovement'[Speed_Category] = "Slow"
)`

Explanation:

DISTINCTCOUNT counts each vessel only once using its MMSI. The CALCULATE function then applies a filter so that only records having the speed category "**Slow**" are considered.

This provides a KPI for identifying the number of unique slow-moving vessels in the dataset.

Used in:

- KPI Card – **Slow-Moving Vessels**

10.4.3 FAST-MOVING VESSELS

Purpose:

This measure calculates the number of unique vessels classified as **Fast**.

DAX Formula:

Fast-Moving Vessels =

```
CALCULATE(  
    DISTINCTCOUNT('Fact_VesselMovement'[MMSI]),  
    'Fact_VesselMovement'[Speed_Category] = "Fast"  
)
```

Explanation:

The measure uses DISTINCTCOUNT on the MMSI field to count unique vessels. CALCULATE filters the data so that only vessels with the **"Fast"** speed category are included.

This helps provide a quick view of the number of fast-moving vessels represented in the dataset.

Used in:

- KPI Card – **Fast-Moving Vessels**

10.4.4 ETA VARIABILITY (HRS)

Purpose:

This measure calculates the variability of the ETA values in hours.

DAX Formula:

```
ETA Variability (hrs) =  
STDEV.P('Fact_VesselMovement'[ETA_hours])
```

Explanation:

The STDEV.P function calculates the **population standard deviation** of the ETA_hours values. A higher value indicates greater variation in ETA values, while a lower value indicates that the ETA values are more consistent.

This measure was used to provide a high-level view of ETA variability in the dashboard.

Used in:

- KPI Card – **ETA Variability (hrs)**

10.4.5 ETA VARIANCE (HRS)

Purpose:

This measure calculates the difference between the current average ETA and the overall average ETA across all destination clusters.

DAX Formula:

ETA Variance (hrs) =

```
AVERAGE('Fact_VesselMovement'[ETA_hours])  
-  
CALCULATE(  
    AVERAGE('Fact_VesselMovement'[ETA_hours]),  
    ALL('Fact_VesselMovement'[dest_cluster])  
)
```

Explanation:

The first AVERAGE calculates the average ETA in the current filter context, such as a particular destination cluster.

The CALCULATE function calculates the overall average ETA while ALL('Fact_VesselMovement'[dest_cluster]) removes the destination-cluster filter.

The two values are then subtracted to determine how much the selected destination's average ETA differs from the overall average.

- **Positive value:** Average ETA is higher than the overall average.
- **Negative value:** Average ETA is lower than the overall average.
- **Value near zero:** Average ETA is close to the overall average.

Used in:

- **ETA Variance by Destination** chart

10.5. DASHBOARD DEVELOPMENT

For Milestone 2, an interactive Power BI dashboard was developed to provide deeper analysis of vessel movement and operational performance. The dashboard builds upon the data model and measures developed in the previous milestone and presents the results through KPI cards, charts, drill-down analysis, and interactive filtering.

The dashboard focuses on distance, vessel movement, destination, ETA, cargo, and variance analysis. The visuals were designed to provide both high-level performance indicators and detailed analytical insights while maintaining a clear and user-friendly layout.

Then we'll break the dashboard explanation into **four subsections**:

10.5.1 KPI ANALYSIS

Four KPI cards were developed to provide a quick overview of important operational indicators in the vessel movement dataset. These KPIs allow users to understand the overall movement distance, ETA variability, and vessel speed distribution at a glance.

1. ETA Variability (hrs)

The ETA Variability KPI represents the variation in estimated arrival times across the available vessel movement records. It is calculated using the population standard deviation of ETA_hours. This helps indicate how consistent or variable the ETA values are across the dataset.

2. Total Distance Travelled (km)

The Total Distance Travelled KPI shows the cumulative recorded distance covered by vessel movement records. It provides a high-level view of the overall movement distance represented in the dataset.

3. Slow-Moving Vessels

The Slow-Moving Vessels KPI represents the number of unique vessels classified under the Slow speed category. It helps identify the portion of vessel activity associated with slower movement.

4. Fast-Moving Vessels

The Fast-Moving Vessels KPI represents the number of unique vessels classified under the Fast speed category. It provides a quick indication of vessels operating at higher movement speeds.

10.5.2 DESTINATION & MOVEMENT ANALYSIS

This section explains the visuals used to analyze destination-wise vessel activity, movement patterns over time, and ETA performance.

1. Total Distance by Destination

The Total Distance by Destination chart compares the total recorded vessel movement distance across different destination clusters. It helps identify destinations associated with higher or lower overall vessel movement distance and provides a clearer view of destination-wise operational activity.

2. Vessel Movement Trend by Hour

The Vessel Movement Trend by Hour line chart shows how vessel movement activity changes throughout the day. The hourly analysis helps identify periods of relatively higher and lower vessel activity, making it useful for understanding the time-based pattern of vessel movements.

3. Average ETA by Destination

The Average ETA by Destination chart compares the average estimated arrival time across destination clusters. It helps identify destinations with relatively higher or lower average ETA values and provides a basis for comparing arrival-time performance between different destinations.

10.5.3 TREND, VARIANCE & RELATIONSHIP ANALYSIS

This part of the dashboard focuses on identifying differences in ETA performance and understanding the relationship between distance, ETA, and vessel speed.

1. ETA Variance by Destination

The ETA Variance by Destination chart compares the average ETA of each destination cluster with the overall average ETA. The variance measure helps identify destinations where the average ETA differs from the overall benchmark.

- A **positive variance** indicates that the destination's average ETA is higher than the overall average.
- A **negative variance** indicates that the destination's average ETA is lower than the overall average.
- A value **close to zero** indicates that the destination's average ETA is close to the overall average.

This visualization helps users identify destinations with noticeable deviations in ETA performance.

2. Distance vs ETA Analysis

The Distance vs ETA Analysis scatter chart is used to examine the relationship between vessel movement distance and estimated arrival time across different speed categories.

The visual helps in understanding whether vessels associated with greater distances also tend to have higher ETA values, while the speed-category breakdown provides additional context about vessel movement behavior.

10.5.4 CARGO & DRILL-DOWN ANALYSIS

This section focuses on cargo-wise movement analysis and detailed vessel-level exploration. It also includes the interactive destination filter used to make the dashboard more dynamic.

1. Top 10 Cargo Categories by Average Distance

The Top 10 Cargo Categories by Average Distance bar chart compares cargo categories based on their average recorded movement distance. The visual is limited to the top 10 categories to keep the analysis clear and readable.

This helps identify cargo categories associated with relatively higher average movement distances and provides an additional perspective for understanding cargo-related vessel activity.

2. Cargo → Vessel Drill-Down

The Cargo → Vessel Drill-Down chart provides hierarchical analysis by allowing users to move from a high-level Cargo category to individual VesselName records.

This enables users to:

- Analyze cargo categories at a summarized level.
- Select a cargo category for further investigation.
- Drill down to the vessels associated with the selected cargo.
- Examine vessel-level details within a particular cargo category.

This fulfills the drill-down analysis requirement of Milestone 2 while providing more detailed exploration of the available data.

3. Destination Cluster Filter

A Destination Cluster dropdown slicer was added to the dashboard to provide interactive filtering. Users can select a particular destination cluster and dynamically update the dashboard visuals according to the selected destination.

This allows the analysis to be performed both at an overall level and for a specific destination cluster.

